

Research Experience

- Research Group** 📖 Working as a PhD student at the EDGES lab, UIC.
- Research Interests** 📖 Causal Inference, Causal Discovery, Time-series Analysis, Graph Neural Networks, Large Language Models.
- Description:** My interest lies in the intersection of causal inference, data science, and machine learning. I am particularly fascinated by social network graphs recently. I want to go beyond machine learning to explainability, unlearning, and causality.

Research Publications

Journal Articles

- 1 S. S. Arnob, **A. A. Shuvro**, S. Rahman, M. Moniruzzaman, and M. S. Hossen, "Time-series multivariate multistep traffic flow forecasting using temporal fusion transformers," *International Journal of Intelligent Transportation Systems Research*, pp. 1–7, 2025. 🔗 DOI: 10.1007/s13177-025-00480-1.
- 2 **A. A. Shuvro**, M. S. I. Bhuiyan, F. Hussain, and M. S. Hossen, "Zero-shot performance analysis of large language models in sumrate maximization," *PLoS One*, vol. 20, no. 8, e0329674, 2025. 🔗 DOI: 10.1371/journal.pone.0329674.
- 3 **A. A. Shuvro**, M. S. Khan, M. Rahman, F. Hussain, M. Moniruzzaman, and M. S. Hossen, "Transformer based traffic flow forecasting in sdn-vanet," *IEEE Access*, vol. 11, pp. 41 816–41 826, 2023. 🔗 DOI: 10.1109/ACCESS.2023.3270889.

Conference Proceedings

- 1 S. A. Chowdhury, S. F. Karim, M. S. I. Bhuiyan, **A. A. Shuvro**, and M. S. Hossen, "Quantum machine learning for anomaly detection in iot devices," in *2025 5th International Conference on AI-ML-Systems (AIMLSystems)*, 2025, pp. 1–6. 🔗 DOI: 10.1109/AIMLSystems67835.2025.11331191.
- 2 A. A. Arnab, K. Ma, **A. A. Shuvro**, and H. Leung, "Comparison of 4g lte and 5g nr in uav networks: A simu5g-based performance evaluation," in *2023 IEEE 9th World Forum on Internet of Things (WF-IoT)*, 2023, pp. 1–6. 🔗 DOI: 10.1109/WF-IoT58464.2023.10539559.
- 3 A. A. Arnab, **A. A. Shuvro**, K. Ma, and H. Leung, "A deep learning approach for a qos prediction system in cellular networks," in *2023 IEEE 9th World Forum on Internet of Things (WF-IoT)*, 2023, pp. 1–6. 🔗 DOI: 10.1109/WF-IoT58464.2023.10539507.
- 4 R. M. Alamgir et al., "Performance analysis of yolo-based architectures for vehicle detection from traffic images in bangladesh," in *2022 25th International Conference on Computer and Information Technology (ICCIT)*, 2022, pp. 982–987. 🔗 DOI: 10.1109/ICCIT57492.2022.10055758.

Ongoing Research

- 1 **A. A. Shuvro**, D. Arbour, and E. Zheleva, *Enhanced LiNGAM-based Causal Discovery for Relational Data in Networks*.

Employment History

- 08/2024 – Present  **Graduate RA/TA (PhD student), University of Illinois Chicago**
Department of Computer Science
Current Research: Causal Discovery for Relational Data in Networks
Courses Taken: Causal Inference and Learning, Machine Learning in Graphs, Research Methods in CS, Introduction to Machine Learning, Introduction to Data Science, Adv. Topics in Econ. & Computing (Game Theory), Computer Algorithms.
- 08/2022 – 08/2024  **Lecturer, Islamic University of Technology**
Department of Computer Science and Engineering
Courses conducted: Cryptography and Network Security, Computer Networks Lab, Software Project Lab, Bioinformatics Lab, Computer Science and Technology Lab, Compiler Design Lab, Numerical Methods Lab, Design Project, Engineering Ethics, Computer Programming Lab, Data Structure and Algorithms Lab, and E-Commerce and Web Security.
- 09/2022 – 05/2023  **Contractual Lecturer, Brac University**
Department of Computer Science and Engineering
Courses conducted: Algorithms Lab, Computer Networks Lab, System Analysis and Design Lab, Robotics Lab, Compiler Design Lab.
- 04/2021 – 09/2021  **Machine Learning Engineer Intern, Pioneer Alpha Ltd.**
Worked with the Vision Research Group in creating image generative projects by using computer vision models.
- 02/2021 – 04/2021  **Network Service Management Intern, Banglalink Digital Communication**
- Worked with the Network Service Management team to gather knowledge about the backbone of real-life networks.
- Collaborated with the managers and engineers of Nokia to implement the Zero-touch configuration on base stations.

Education

- 2024 – Present  **PhD Computer Science, University of Illinois Chicago (Ongoing)**
Current CGPA: 4.00 (out of 4.00)
- 2018 – 2022  **B.Sc. Computer Science, Islamic University of Technology**
CGPA: 3.97 (out of 4.00) - *Top of the class (Gold Medalist)*
Thesis title: *Time-series traffic flow forecasting in Software Defined Networks using Transformer-based model.*

Undergraduate Software Projects

- IUT E-Health System**  A cross-platform software implementing E-health system of IUT. The desktop application is developed using JavaFx and the android application is developed using Android Studio. Firebase is used as the shared database.
- Virtual world**  Different objects like coffeeshop, cars, streets and trees were designed using Blender.
- Cocktail Party**  A multi-voice separation technique for separating English, Bengali and Spanish. The project is implemented using MATLAB.
- Hoopla**  A desktop social networking application with some basic features. The project is developed using QtCreator in C++.
- Sudoku**  A Sudoku console game is implemented using C.

Skills

Languages	Strong reading, writing and speaking competencies in English and Bengali.
Programming Languages	C, C++, Java, Python, R, $\text{\LaTeX}$, ...
Databases	MySQL, ORACLE SQL.
IDE Proficiency	CodeBlocks, Google Colab, Visual Studio Code, IntelliJ, Eclipse.
Misc.	Academic research, teaching, training, consultation, $\text{\LaTeX}$ typesetting and publishing.

Miscellaneous Experience

Academic Honors and Scholarships

2022	Gold Medalist , 34 th Convocation of Islamic University of Technology.
01/2018 – 05/2022	OIC Scholarship holder , Islamic University of Technology OIC Academic Scholarship.
2009	Talentpool Scholarship holder , Primary School Completion Examination.

Awards and Achievements

2022	Finalist , Robi Datathon 2.0.
2021	National Finalist , Huawei ICT Competition.
	Finalist , Blockchain Olympiad Bangladesh.
2020	Finalist , Dhaka-AI 2020.

Extra-Curricular Activities

2022-2024	Advisor , IUT Computer Society.
	Member , IUT CSE Course Load Distribution Committee.
2022-2023	Programming Coach , National Girls' Programming Contest.
2021-2022	Vice-Captain and Central Midfielder , IUT CSE Football Team.

References

Elena Zheleva, Ph.D.

Associate Professor

University of Illinois Chicago,

✉ ezheleva@uic.edu